

Problem Statement and MCTS Justification

Problem Domain

The Snake game is a sequential decision-making problem in which an autonomous agent must repeatedly choose actions in a dynamic environment to maximize long-term reward.

At each timestep, the agent selects one of four possible actions:

- Move Up
- Move Down
- Move Left
- Move Right

The challenge is that locally optimal actions do not always produce globally optimal outcomes.

For example, moving directly toward food may increase immediate reward but can trap the snake in a dead-end path later, resulting in collision and game termination.

This makes Snake a planning-intensive search problem rather than a simple reactive control task.

The AI must reason about:

- Immediate reward acquisition
- Long-term survivability
- Collision avoidance
- Future board state consequences

The objective is to maximize score while maintaining safe movement over time.

The computational challenge arises because the game contains:

- Sequential action dependency
- Dynamic state transitions
- Delayed consequences
- Large search complexity over extended play

Exhaustive search becomes computationally infeasible as possible future trajectories grow exponentially.

Why Monte Carlo Tree Search is a Perfect Fit

Monte Carlo Tree Search (MCTS) is highly suitable for Snake because it evaluates actions through repeated simulation rather than brute-force enumeration.

MCTS is a strong fit because:

1. Online Decision Making

The algorithm evaluates moves during gameplay.

2. Exploration vs Exploitation Balance

It explores new paths while refining successful strategies.

3. Adaptability

It dynamically responds to changing board configurations.

4. No Handcrafted Heuristics Required

The algorithm learns move quality through simulation.

5. Anytime Performance

Additional iterations improve decision quality.

By combining simulation-based reasoning with statistical evaluation, MCTS provides an effective solution for autonomous Snake gameplay.